

MITCHELL, SD, USA

WMO: 726545

Lat: 43.774N

Lon: 98.038W

Elev: 1300

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 94950

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.1	-5.6	-18.5	2.1	-8.9	-12.9	2.9	-4.4	34.9	10.8	31.1	18.4	9.9	330	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	93.8	74.1	90.6	73.1	87.9	71.7	77.6	88.6	75.8	86.6	74.1	84.6	12.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.4	134.8	84.2	72.5	126.2	82.6	70.5	118.0	80.7	42.0	88.1	40.2	86.2	38.5	84.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.4	25.1	22.3	-17.3	99.7	7.0	4.6	-22.3	103.0	-26.4	105.7	-30.4	108.2	-35.4	111.6
			-16.9	80.5	5.9	2.4	-21.2	82.2	-24.6	83.6	-28.0	85.0	-32.3	86.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	18.6	22.7	34.2	47.2	59.2	69.7	75.3	72.0	64.2	50.2	35.4	22.5
	DBStd	22.25	13.27	13.31	13.13	10.17	8.72	6.94	6.35	5.65	8.58	9.61	11.21	11.98
	HDD50	3860	975	765	507	169	19	0	0	0	5	117	450	853
	HDD65	7179	1440	1184	954	538	223	33	6	9	118	468	888	1318
	CDD50	3026	0	0	19	84	304	590	783	683	430	122	11	0
	CDD65	871	0	0	0	4	43	172	324	227	93	8	0	0
	CDH74	9106	0	0	13	100	522	1735	3493	2110	1004	125	4	0
	CDH80	3569	0	0	1	23	163	637	1576	772	363	34	0	0
(14) Wind	WSAvg	10.5	11.0	11.2	11.7	12.4	11.3	9.9	8.8	8.6	9.6	10.7	10.7	10.7
(15) Precipitation	PrecAvg	21.9	0.4	0.6	1.7	2.5	3.2	3.7	2.5	2.2	2.4	1.5	0.9	0.6
	PrecMax	32.6	1.5	2.9	7.5	7.1	10.0	11.1	5.3	6.6	6.9	4.6	3.7	2.0
	PrecMin	13.0	0.0	0.0	0.2	0.4	0.7	0.4	0.3	0.3	0.1	0.0	0.0	0.0
	PrecStd	5.2	0.4	0.7	1.5	1.5	2.2	2.2	1.4	1.3	1.5	1.3	0.9	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.2	60.5	75.1	82.7	90.9	95.2	101.3	96.4	93.0	84.3	71.5	54.6
		MCWB	43.2	47.8	55.5	61.0	68.2	74.6	74.8	75.1	70.4	62.8	54.1	43.3
	2%	DB	44.6	51.8	66.4	76.0	84.1	90.4	95.8	90.8	88.6	76.9	63.6	47.2
		MCWB	37.8	43.0	52.0	57.3	65.4	72.1	75.2	74.4	69.6	60.0	50.7	39.5
	5%	DB	39.3	45.3	59.4	70.3	80.1	87.0	91.4	87.7	84.0	71.6	56.7	41.8
		MCWB	35.0	39.1	48.7	54.3	63.2	70.6	74.8	72.6	67.9	57.6	47.0	35.8
	10%	DB	36.3	39.9	53.6	64.8	75.1	83.4	88.4	84.0	80.3	65.8	51.7	37.2
		MCWB	33.1	35.3	45.2	51.6	60.8	68.6	73.2	71.1	65.9	54.5	43.3	33.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	49.7	59.0	63.5	71.2	78.0	80.4	79.2	74.7	66.5	56.5	45.4
		MCDB	53.2	60.3	67.4	79.7	85.3	90.4	91.5	90.2	85.4	78.5	67.0	52.8
	2%	WB	38.3	43.0	54.2	59.3	68.1	75.0	78.2	76.8	72.4	62.6	51.5	39.8
		MCDB	43.3	50.2	64.2	72.8	80.2	86.8	89.4	87.0	84.1	72.3	61.6	46.6
	5%	WB	35.3	39.3	49.9	55.8	65.4	72.7	76.6	75.0	70.4	59.1	47.5	36.5
		MCDB	39.1	45.1	59.9	67.7	77.4	83.5	87.7	84.5	81.3	69.4	56.7	41.3
	10%	WB	33.1	35.6	45.0	53.0	62.6	70.5	74.8	73.0	68.0	55.5	43.6	33.7
		MCDB	36.2	40.0	52.7	63.1	72.7	80.8	85.8	81.7	77.1	64.6	51.3	37.1
(35) Mean Daily Temperature Range	5% DB	MDBR	17.2	18.6	19.6	21.8	21.2	20.6	21.9	21.5	23.6	21.7	20.0	16.4
		MCDBR	22.0	25.1	30.9	32.1	28.1	24.8	25.6	25.4	28.8	30.4	29.5	22.9
	5% WB	MCWBR	16.8	17.4	18.6	16.8	13.7	11.3	10.5	11.7	13.1	16.1	18.0	16.5
		MCWBR	20.9	24.0	30.1	28.1	25.1	22.2	22.6	21.7	24.3	26.4	28.8	22.0
(40) Clear-Sky Solar Irradiance	taub		0.269	0.287	0.313	0.364	0.384	0.387	0.395	0.399	0.351	0.318	0.288	0.265
		taud	2.449	2.424	2.436	2.301	2.331	2.372	2.358	2.328	2.438	2.499	2.493	2.472
	Ebn at Noon		277	290	295	285	282	280	276	271	278	275	267	268
		Edh at Noon	23	28	31	39	39	38	38	38	31	26	22	20
(44) All-Sky Solar Radiation	RadAvg		601	927	1224	1551	1819	2043	2148	1785	1420	955	650	480
	RadStd		51	65	107	130	144	147	71	135	104	110	59	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page